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(Approved by AICTE, New Delhi & Affiliated to JNTUK, Kakinada)

An ISO 9001:2015 Certified Institute

Department Of Electronics & Communications Engineering

Subject Name: BioMedical Instrumentation

Unit No: 3

Patient Care & Monitoring ; Therapeutic & Prosthetic Devices

- 4.1 Elements of Intensive Care Monitoring
- 4.2 Patient Monitoring Displays
- 4.3 Diagnosis, Calibration & Repairability of PME.
- 4.4 Other Instrumentation for Monitoring Patients.
- 4.5 Organization of hospitals for Patient Care Monitoring.
- 4.6 PaceMakers
- 4.7 Defibrillators.

Unit - 4

Patient Care And Monitoring

4.1 Elements of Intensive - Care Monitoring :-

The basic elements of Intensive Care Monitoring

unit are

- (i) Skin electrodes to pickup ECG potentials.
- (ii) Amplification unit for Electro Cardiograph.
- (iii) CRT display for observation of ECG waveforms.
- (iv) A rate meter to indicate heart beat.
- (v) An alarm system which is actuated by the rate meter to alert the nurse.
- (vi) A direct wired up device for Electro cardiograph.
- (vii) A magnetic tape to record the playback of ECG for 15 to 60 Sec.
- (viii) Additional alarm system triggered by ECG parameters.
- (ix) Electrical circuit to indicate that electrode connections have been disconnected due to mechanical failure.
- (x) A Closed circuit Monitoring System is also employed to provide visual Coverage using Cameras focused on each patient.

Patient - Monitoring equipments are generally specified as a system. Each manufacturer and each hospital staff has its own ideas. A wide variety of configurations can be seen in the hospitals.

A Simple patient monitoring System is as shown below

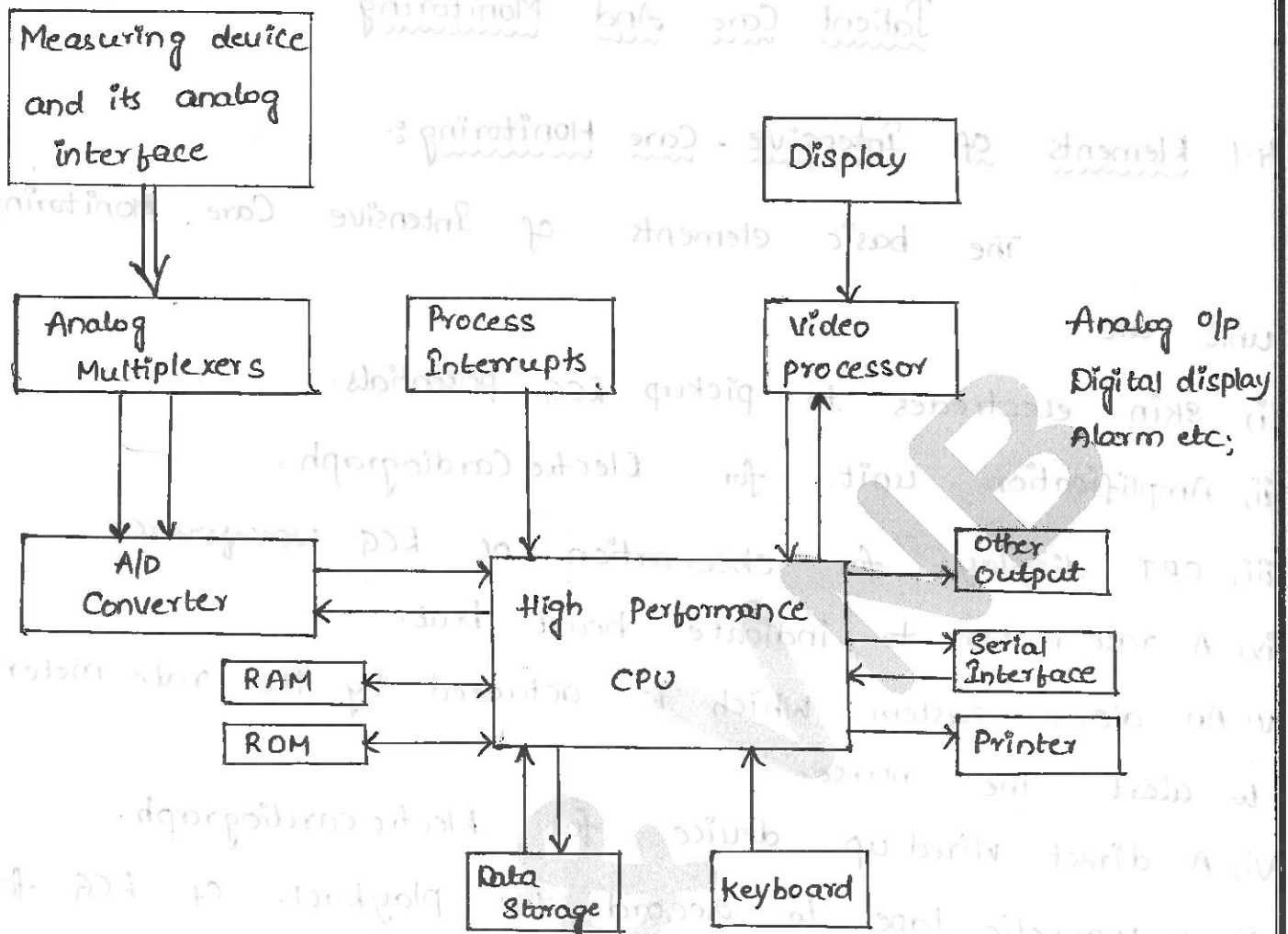


fig:- Patient Care Monitoring System Block diagram

The communication channel from the Central Station to each bedside is bidirectional So that data and Control Signals can flow freely in each direction between Micro Computers.

The patient care Stations accept data in general upto eight patients. It may provide Single or Dual channel Strip chart recordings.

A.2 Patient Monitoring Displays :-

Faithful reproduction of ECG, Arterial Blood pressure and other valuable parameters are essential.

There are three types of displays.

→ Conventional Display :- This uses conventional CRT with horizontal sweep driven by a slow-speed sweep generator which causes the electron beam to move from left to right at a predetermined rate i.e. 25 to 50 mm/sec.

→ Bouncing Ball Display :- This is also CRT arrangement where sweep rate is very high. Each spot of light seems to be bouncing up, hence the name bouncing ball display.

→ Non fade display :- It also uses CRT display, but the electron beam rapidly scans the entire screen of the CRT in a television like raster pattern.



fig:- ECG pattern made of dots

The beam is brightened only when a brightening signal is applied to CRT by x-axis modulation. This gives trace pattern i.e. a series of dots on the screen.

4.3 Diagnosis, Calibration and Repairability of Patient Monitoring

Equipment :-

- Engineers and Technicians need to diagnose the problems of monitoring equipments.
- It is essential that equipments are operational and calibrated accurately.
- The spread of medical instruments with high degree of complexity has created a great demand for effective and efficient trouble shooting and maintenance.

There are several major activities involved in the process of installation of a large expensive biomedical instrumentation such as X-ray, CT Scan, MRI etc. The installation must be well planned with each detail.

The manuals must be obtained for installation, operation and maintenance. Spare parts must be provided as backup in case of failures.

- Environment conditions of the instrumentation would generally be good, however temperature, dust, humidity should be in tolerance limit.
- No heavy machinery should be around those generate and transmit vibrations.
- It should have protection against heavy intermittent currents.

The electrical power supply should be stable

Balanced and Preferably draws from Connected Secondary of the transformer. To minimize the effects electrical grounding should be provided for each instrumentation System.

The basic electrical equipments required for installation and maintenance are;

- Multimeter
- Oscilloscope
- Temperature Controlled Soldering Equipments.
- Extended cards & Spares.
- Logic probes.

There are some Offline diagnostics which guide the maintenance engineer through Step-by-step process analysis of the instrumentation System.

- CPU Diagnostics
- Memory Diagnostics
- CRT Diagnostics
- D/A and A/D Diagnostics
- I/O Diagnostics.

Sometimes the biomedical instrumentation Suffer when Subjected to excessive vibration;

- Friction Connections like circuit board socket or plug socket can move.
- circuit boards can gradually vibrate out of their sockets

→ Lead Connections Can be Shivered.

The remedy against vibration is to place vibration - absorbing mounting pads, or to use rubber - center vibration isolating bolt mounts.

Power - Supply Conditioning :-

To ensure instrumentation system reliability is to protect them with new generation solid - state transient suppressors.

Without such suppressors, transient noise or voltage spikes can enter into system via power supply or signal lines and cause disruption & destruction.

Even if power supplied to the hospital is relatively clean, voltage spikes and other transients are generated within the hospital itself; as a result of operating monitors, fluorescent lights, switching circuits etc.

If these transients get into computerized equipment, they may be misinterpreted as digital commands. Therefore they must be avoided. The devices used for this purpose are;

→ Inductive - Capacitive filters.

→ Resistance - Capacitance filters.

→ Metal oxide Varistor.

→ Isolation Transformer.

4.4 Other Instrumentation for Monitoring Patients :-

There are several other special medical instrumentation in use which are other than bedside of the patient. They are :

1. Surgical Monitoring System :- The main features of this type of system are the large multichannel oscilloscope, the capability of obtaining a permanent ECG record on the chart recorder, and plug-in signal conditioner modules that provide choice of measurement parameters.
2. Arterial Diagnostic Unit (ADU) :- This mobile cart contains several commonly used instruments for peripheral arterial evaluation. The ADU provides automated pressure-cuff inflation for rapidly determining segmental pressures in arterial evaluations. It has a heated stylus strip-chart recorder, a bidirectional doppler-flow meter with external loud speaker and headphones.
3. Catheterization Lab :- It is a lab where cardiologists perform diagnostic catheterization. For a patient having blockage in one of the coronary arteries, must be brought into the cath lab.
Catheterization is a dangerous procedure, therefore patient must be monitored continuously. Many units have extensive built-in monitoring systems.

4.5 Organization of the Hospital for patient Care

Monitoring :-

The engineer or technician should be familiar with the overall organization of the hospital with respect to monitoring equipment.

Generally the equipments are divided into two types.

(i) Surgical Equipments :- These are placed in the operating room, where surgery is actually performed.

(ii) Non-Surgical Equipments :- The monitoring equipments, made available in operation room are heart rate meter, Venous and arterial blood pressures, ECG, EEG, Various respiratory therapy devices.

It may also have emergency equipments like Defibrillators and Pacemakers, Stimulation equipments etc; The Intensive Care unit can be used for post-surgical follow-up with very serious problems.

It provided with equipment similar to that of the operating room, the only difference is usually only one set of equipment in the operating room, while in the ICU there are bedside monitoring units and nurses.

As the heart attack patient recovers, he or she is usually moved into another unit called as Inter-mediate Coronary care unit [ICCU]; may contain telemetry.

4.6 Pacemakers :-

A Pacemaker is a small device that is placed under the skin in our chest to help control our heart beat. It is used to help our heart to beat more regularly if we have an irregular heart beat. ~~Many~~ Implanting of pacemaker in the chest requires a surgical procedure.

Types of Pacemakers :-

Depending on the condition of the patient's heart;

One of the following types of pacemakers are used.

- Single chamber Pacemaker :- This type of pacemaker usually carries electrical impulses to the right Ventricle of heart.
- Dual chamber Pacemaker :- This type of pacemaker carries the electrical impulses to the right Ventricle and the right atrium of the heart. It controls the timing of contractions between the two chambers.
- Biventricular Pacemaker :- It is also called Cardiac Resynchronization therapy; used for the people with heart failure with abnormal electrical systems. This type of pacemaker stimulates the lower chambers of heart to make the heart beat more efficiently.

Working of a Pacemaker :-

An implanted electronic pacemaker mimics the action of natural electrical system.

A pacemaker comprises of two parts.

→ Pulse Generator :- This contains a battery and the electrical circuitry that regulates the rate of electrical pulses sent to the heart.

→ Leads [Electrodes] :- One to three flexible insulated wires are placed in chambers of heart to deliver the electrical pulses to adjust heart rate.

Pacemakers work only when needed. If the heart beat is too slow, the pacemaker sends electrical signals to heart to correct the beat.

Newer pacemakers have sensors that detect body motion or breathing rate, and alert the pacemakers to increase heart rate during exercise, as needed.

Too smaller, leadless pacemakers, which can be implanted directly into the heart, have been approved for use in United States.

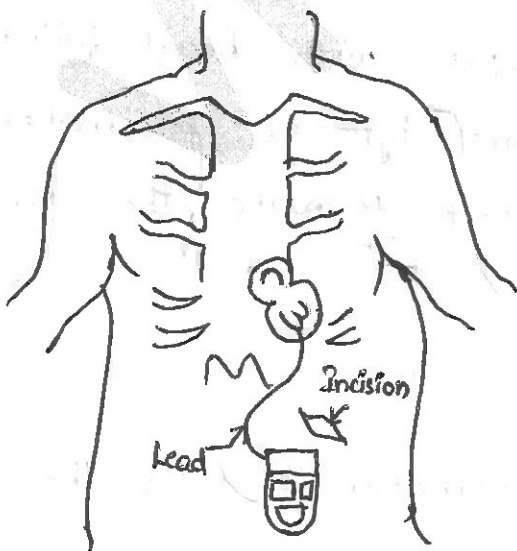


fig:- Pacemaker Generator implanted in abdomen

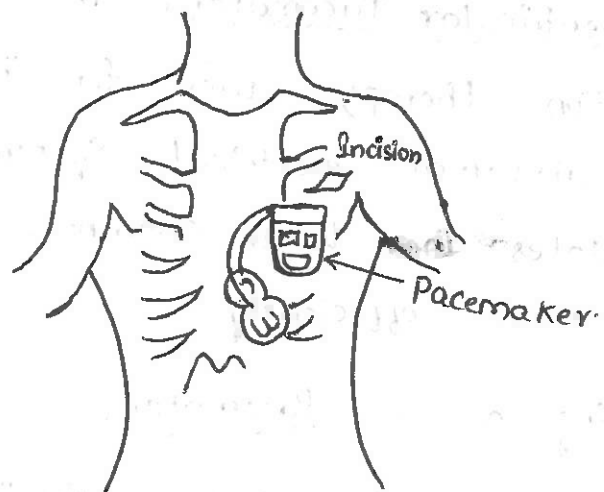
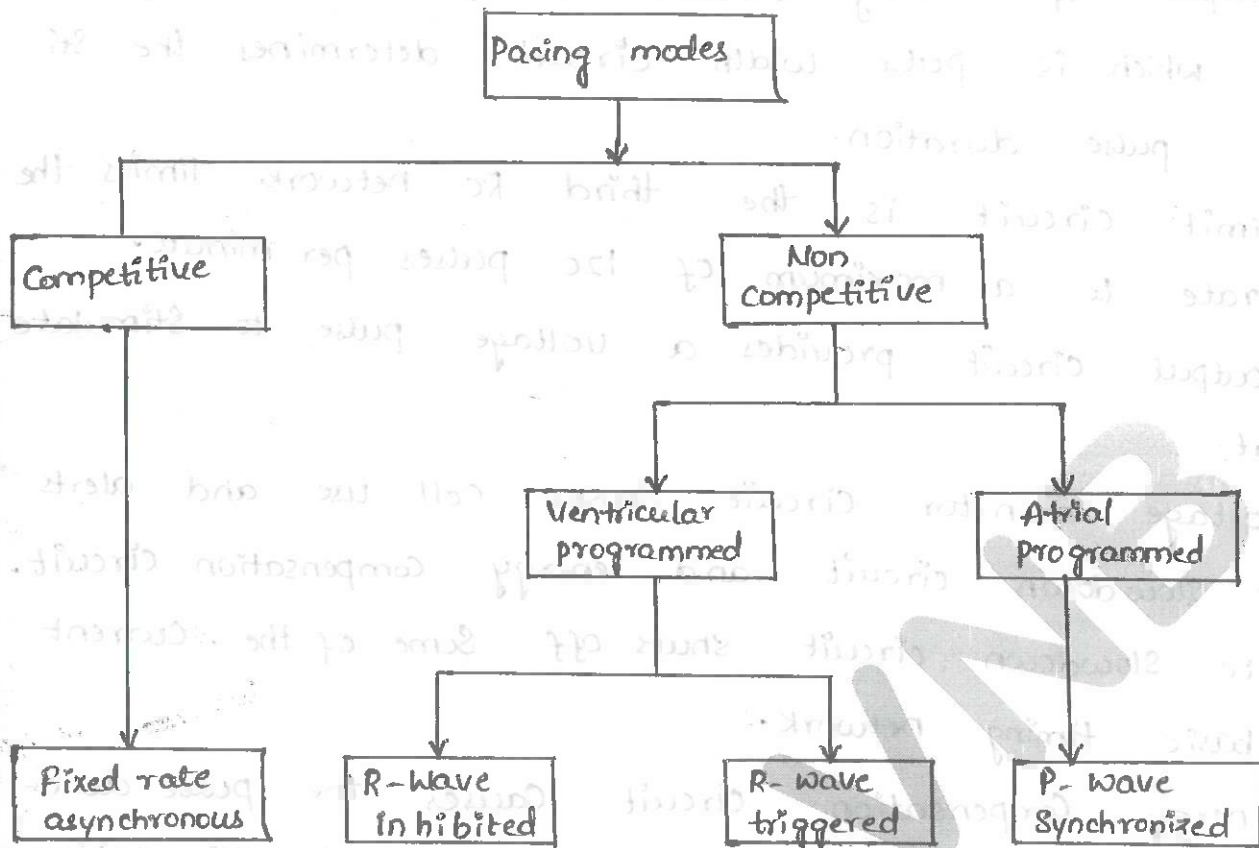
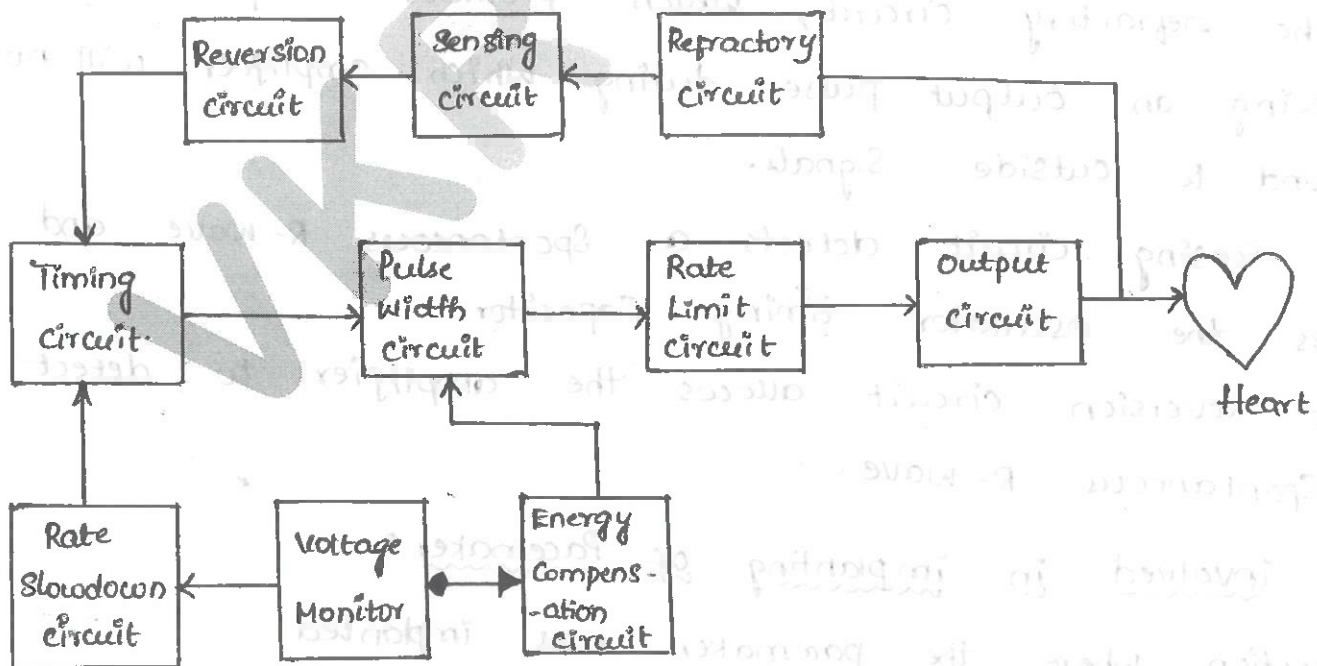


fig:- Implanted Pacemaker with electrodes catheter

Types of Pacing modes :-



Internal Circuitry of Pacemaker :-



→ The timing circuit which consists of an RC network, a reference voltage source, and a comparator determines the

basic pacing rate of the pulse Generator.

→ The output of timing circuit is fed into Second RC network, which is pulse width circuit determines the stimulating pulse duration.

→ Rate limit circuit is the third RC network limits the pacing rate to a maximum of 120 pulses per minute.

→ The output circuit provides a voltage pulse to stimulate the heart.

→ The voltage monitor circuit senses cell use and alerts the rate slowdown circuit and energy compensation circuit.

→ The Rate slowdown circuit shuts off some of the current to the basic timing network.

→ The Energy compensation circuit causes the pulse duration to increase to maintain nearly constant stimulation energy to the heart.

→ There is also a feedback loop from the output circuit to the refractory circuit, which provides a period of time following an output pulse during which amplifier will not respond to outside signals.

→ The Sensing circuit detects a spontaneous R-wave and resets the oscillator timing capacitor.

→ The Reversion circuit allows the amplifier to detect a spontaneous R-wave.

Risks involved in implanting of Pacemaker:-

→ Infection where the pacemaker was implanted

→ Swelling, Bleeding at the generator site

→ Damage to blood vessels & collapsed Lung.

4.7 Defibrillators :-

Defibrillators are the devices that restore a normal heart beat by sending an electric pulse or shock to the heart.

They are used to prevent or correct Arrhythmia, a heart beat that is uneven or that is too slow or too fast.

Defibrillators can also restore the heart's beating if the heart suddenly stops.

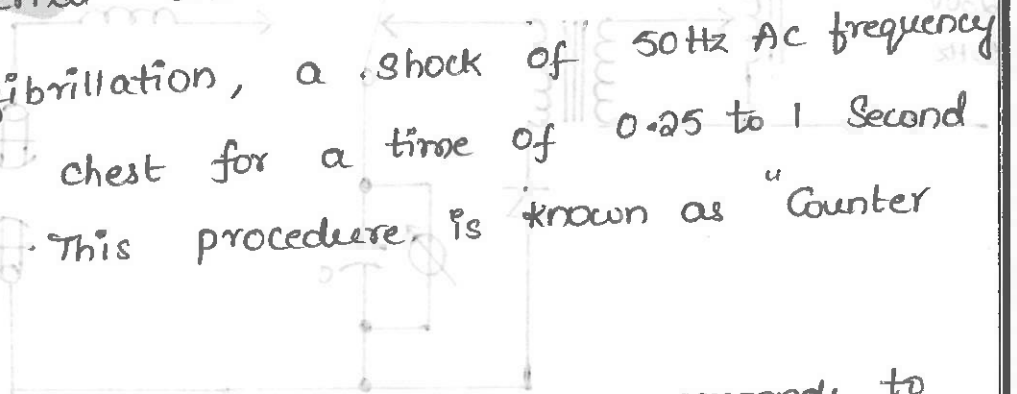
Types of Defibrillators :-

1. AC Defibrillators
2. DC Defibrillators.

AC Defibrillators :-

It is the oldest and simplest type. The construction of AC defibrillator is such that appropriate values are available for internal and external defibrillation.

In AC defibrillation, a shock of 50 Hz AC frequency is applied to the chest for a time of 0.25 to 1 second through electrodes. This procedure is known as "Counter Shock".



Defibrillation continues until patient responds to the treatment. However this results in skin burn under electrodes and violent contraction of heart muscles.

DC Defibrillators :-

DC Defibrillator does not produce side effects and produces normal heart beat. Ventricular fibrillation is avoided when high-energy shock is passed through the discharging capacitor that is exposed to heart or chest of the patient.

DC Defibrillator consists of Auto transformer T_1 , that acts as primary of the high voltage transformer T_2 . A diode rectifier rectifies the output voltage from T_2 . A Capacitor is charged to a high dc voltage which is rapidly discharged through electrodes across the chest of the patient.

The DC method requires less repetitions and is less harmful to the patient.

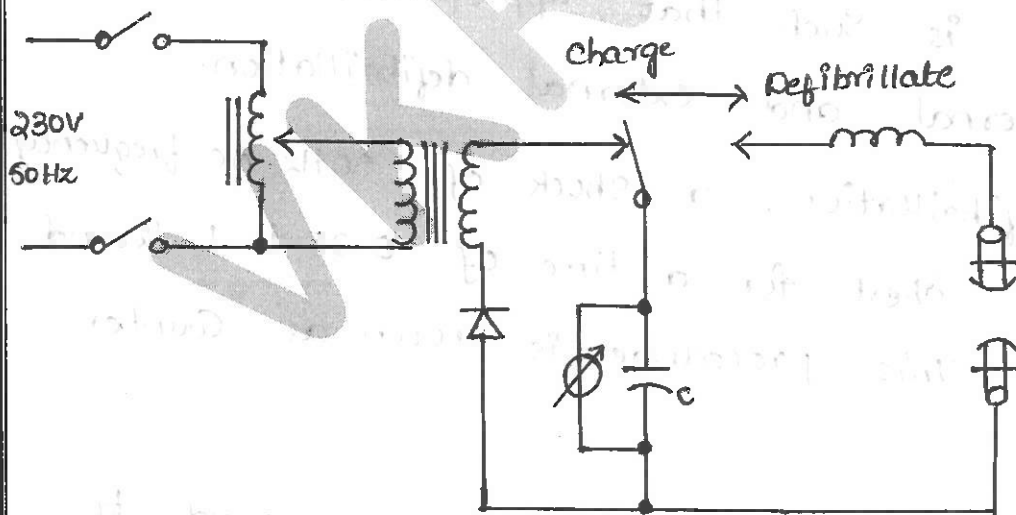


fig:- Circuit of
DC defibrillator

The electrical energy discharged by the Capacitor may vary between 100 to 400 W-Sec or Joules and duration of the effective period of discharge may be about 5mSec.